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A Deep Reinforcement Learning-based Robot Navigation Method using Sensor Information Fusion

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***Abstract*—This paper proposes a Deep Reinforcement Learning (DRL)-based method using multi-sensor information fusion for mobile robot navigation tasks. In order to leverage the precise distance data of the 2D LiDAR and the rich information of the camera image, the proposed method improves the ability of the LiDAR to identify obstacles by leveraging the obstacle information in the camera image, while ensuring that there is no coupling between the image and the LiDAR data. Furthermore, a self-attention network is introduced to enhance the perception of the current environment. The proposed method is evaluated through simulations and a navigation framework based on DRL, which is designed to test and apply perception effects. The experimental results demonstrate that the proposed method enhances the perception and expression ability of the laser radar to obstacles, thereby effectively improving the performance of the robot to avoid obstacles.**

***Index Terms*—deep reinforcement learning, soft actor-critic, mobile robot navigation, sensor fusion, self attention**

I. Introduction

In recent years, there has been a marked increase in the level of interest directed towards intelligent mobile robots. These robots are currently being utilized in a variety of domains, including catering, logistics, environmental monitoring, and domestic assistance [1]. A mobile robot must possess the capacity to perceive its environment, formulate decisions, and modify its actions accordingly. A substantial body of research has been dedicated to addressing the robot navigation problem in unknown environments, with a focus on localization [2]and path planning [3].

A multitude of methodologies have been proposed for the development of robotic motion planning that are customized to unpredictable environments, where cartographic data is either nonexistent or unreliable. Recent advancements in deep reinforcement learning (DRL) for robotic navigation have enabled autonomous intelligent systems to make highly accurate decisions independent of maps. DRL offers an innovative approach for controlling robots by enabling the streamlining of manual design processes through the mapping of complex sensory data to robotic motion commands [4].

In the context of DRL-based path planning problems, Zhang et al. have introduced a mapless navigation approach based on the D3QN algorithm, using LiDAR and depth cameras as sensor inputs [5]. Concurrently, Cimurs et al. have developed an autonomous navigation system for goal-directed exploration in unknown environments, employing the TD3 algorithm as its foundational framework [6]. Xie et al. integrated multi-sensor data fusion to generate a map of pedestrian obstacles, followed by training using the PPO algorithm [7]. Additionally, extensive research has been conducted on the application of the Soft Actor-Critic (SAC) algorithm to mobile robot control tasks [8], [9]. A significant body of research has incorporated a sensor combination of LIDAR and camera in the perception component, suggesting that this combination yields favorable outcomes in applications and merits further in-depth investigation.

However, because the DRL approach uses sensors for real-time perception, even small perceptual differences can have a significant impact on the decision-making process in ways that cannot be predicted by humans [10], and because of the limitations of simulation software, visual sensor-based navigation systems often experience significant performance degradation when transitioning from a simulated to a real environment. In order to address this challenge, this paper proposes a sensor fusion method that utilizes a single-line 2D LiDAR and a color camera. Furthermore, a DRL-based robot navigation framework has been developed to implement the proposed method in specific navigation tasks. The primary contributions of this paper are as follows:

- A sensor fusion approach is presented, with the objective

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of integrating LiDAR and image information. This approach utilizes image data to enhance the performance of LiDAR in recognizing obstacles, and enables a seamless transition from virtual testing environments to practical, real-world implementations.

- A self-attention network has been integrated into a reinforcement learning network with the objective of enhancing the effectiveness of the robot's perception of the environment. The integration of the self-attention network into the processing of input LiDAR sequences enables the robot to allocate augmented cognitive resources to the focused regions of the immediate environment.
- A series of simulation environments were developed for mobile robot navigation tasks in Gazebo using the Turtlebot3 robot model through the Robot Operating System (ROS), and a reinforcement learning navigation framework was designed for it. Subsequently, the effectiveness of the proposed sensor fusion approach in DRL-based navigation tasks was evaluated by simulation scenarios.

II. Preliminaries

A. Soft Actor-Critic Algorithm

The SAC algorithm is a novel deep reinforcement learning method based on the actor-critic architecture [11]. It is founded on the maximum entropy framework, which permits augmented stochasticity in the policy, thereby elevating the exploration rate of the agent.

The objective function of SAC aims to maximize both the future rewards and the entropy of the policy

$$J_\pi = \sum_{t=0}^{T} \underset{(s_t,a_t)\sim\rho_\pi}{E} \left[r\left(s_t,a_t\right) + \alpha H\left(\pi\left(\cdot|s_t\right)\right)\right] \tag{1}$$

where, ρ_π denote the trajectory distribution induced by the policy π, $H\left(\pi\left(\cdot|s_t\right)\right) = E\left(-log(\pi\left(\cdot|s_t\right))\right)$ represents the entropy of the policy, $\alpha \in \mathbb{R}$ represents the temperature coefficient. The cumulative return of state s_t is denoted by

$$G_t = \sum_{i=t}^{\infty} \gamma^{i-t} \left[r\left(s_i,a_i\right) - \alpha \log\left(\pi\left(a_i|s_i\right)\right)\right] \tag{2}$$

The soft state-action value function, denoted by Q, can be expressed as

$$Q^\pi\left(s_t,a_t\right) = r_t + \gamma \underset{(s_t,a_t)\sim\rho_\pi}{E} \left[G_{t+1}\right] \tag{3}$$

where, $\gamma \in \mathbb{R}$ represents the discount factor. The state-action value function Q represents the expected return value of the current state-action pair (s,a).

III. Methodology

A. The Fusion of LiDAR and Images

In the domain of small mobile robotics, single-line 2D LiDAR and monocular color cameras are the most prevalent sensors employed due to financial constraints. However, these sensors possess inherent limitations in terms of environmental perception. The fundamental principles of reinforcement learning mandate recurrent iterations of the training protocol within a simulated environment, while the material properties, textures, and illumination conditions of objects in simulations frequently exhibit significant discrepancies compared to their real-world counterparts. Consequently, the performance of camera sensors in simulations may deviate from their actual performance in real-world scenarios. Conversely, LiDAR has demonstrated proficiency in acquiring depth information surrounding the robot, exhibiting consistency across diverse environments, rendering it particularly well-suited for the transition from simulations to reality.

Some research has explored the application of deep learning methodologies to obstacle detection by utilizing LiDAR data [12]. However, the incorporation of image data holds the potential to further refine this process. To enhance the robot's environmental perception capabilities, a methodology is presented that integrates image-corrected LiDAR scan data. This technique eliminates the reliance on images captured within simulations, thereby ensuring data consistency between simulated and real environments.

1) The transformation of images and LiDAR projection: Subsequent to the implementation of the coordinate system transformation, the LiDAR scan points can be mapped onto the image plane. Given the fixed installation position of the lidar, the origin of the world coordinate system can be set at this specific point. For a given point P with coordinates $[X_w, Y_w, Z_w]$ in the world coordinate system, it is possible to transform this point into the image coordinate system, thereby obtaining a point p_c with coordinates $[x, y]$. This transformation is facilitated by leveraging the intrinsic and extrinsic parameters of the camera. This point is then transformed into the pixel coordinate system to obtain the point p with coordinates $[u, v]$. The mathematical expression for this transformation is

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & T \\ \vec{0} & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{4}$$

where, $f_x \in \mathbb{R}_+$, $f_y \in \mathbb{R}_+$, $u_0 \in \mathbb{R}_+$, and $v_0 \in \mathbb{R}_+$, are the internal parameters of the camera. Conversely, $R \in \mathbb{R}^{3\times3}$ and $T \in \mathbb{R}^3$ denote the external parameters of the camera. Furthermore, $Z_c \in \mathbb{R}_+$ represents the depth of the point in the camera's coordinate system. The application of the aforementioned equation enables the precise determination of the corresponding pixel location of a LiDAR scanning point.

To reproject the pixel points from the image back into the world coordinate system, it is essential to reverse the previously mentioned transformation. The primary challenge in this process is determining the value of Z_c. The transformation equation (4) can be simplified as

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K(R \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + T) \tag{5}$$

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By placing $[X_w, Y_w, Z_w]$ on the left side, (5) can be modified to

$$\begin{aligned} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} &= R^{-1}(K^{-1}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} Z_c - T) \\ &= \underbrace{R^{-1}K^{-1}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix}}_{Mat_1} Z_c - \underbrace{R^{-1}T}_{Mat_2} \end{aligned} \tag{6}$$

In this context, Mat_1 and Mat_2 are 3x1 vectors. Consequently, the computation of Zc can be derived as

$$Z_c = \frac{Z_w + Mat_2(3,1)}{Mat_1(3,1)} \tag{7}$$

where, $Z_w \in \mathbb{R}_+$ represents the height in the world coordinate system. Given that the location of LiDAR serves as the origin, this value of Z_w is equivalent to 0. By substituting the value of Zc into (6), the corresponding world coordinate can be derived based on the pixel coordinates extracted from the image.

2) The correction of LiDAR data based on the detection of image targets: Following the identification of the corresponding pixel locations of the LiDAR scanning points on the image, it is possible to ascertain whether these LiDAR points reside within the volume occupied by an obstacle. This determination is contingent upon the outcomes generated by applying a target detection model to the image. The YOLOv8 model, which has been shown to surpass the YOLOv5 model in terms of performance, particularly in the detection of small objects, is employed in this study [13].

The target detection model generates a bounding box for obstacle O within the image, denoted by the coordinate set $[x_{min}, x_{max}, y_{min}, y_{max}]$. A LiDAR point is considered to have detected an obstacle if its projection $[u, v]$ onto the image plane lies within the confines of this bounding box, irrespective of whether it physically contacted the obstacle or not. In the case of obstacles identified within the camera's Field of View (FOV), the LiDAR points within the recognition frame are substituted. These points are then substituted with the smallest value detected within the relevant range and subsequently multiplied by a factor that is specified based on the type of obstacle. Typically, this factor is set at 0.9, underscoring the significance of avoiding obstacles.

Subsequently, the corrected LiDAR depth data is disseminated in its native format, thereby replacing the original LiDAR data within the camera's FOV. Fig. 1 provides a visual depiction of the anticipated visual outcome following the integration of LiDAR and image information.

B. Self-attention network

In addition to the proposed sensor fusion method, a self-attention network is introduced in the perception part to enhance the perception ability of the focused content in the input information.

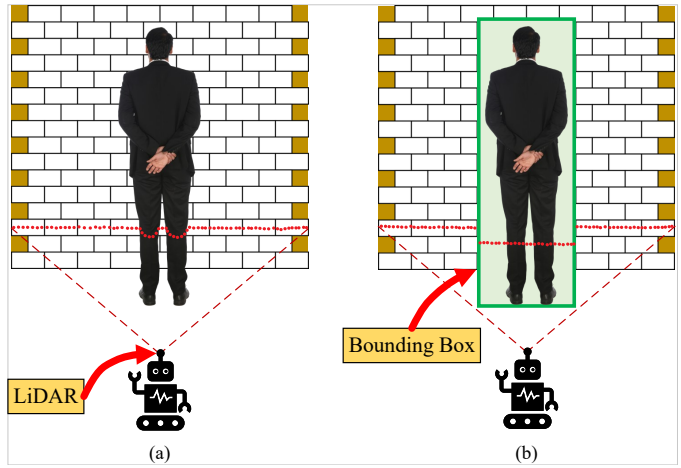

Fig. 1. Schematic diagram of the effect of the sensor fusion method for recognizing obstacles, the red points are the LiDAR scanning points. (a) Projecting effect of raw LiDAR data. (b) Projecting effect of corrected LiDAR data.

The self-attention network computes attentional weights from all elements in the input sequence. These computed weights are subsequently multiplied by the elements to produce a comprehensive weighted-sum output representation. The mechanism of self-attention can be represented as

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{8}$$

where, the symbols Q, K, and V represent the output vectors generated from the processed inputs. In the context of the robot obstacle avoidance navigation task discussed in this paper, the self-attention network is customized to process LiDAR data, encompassing a historical record of prior scans. Through this processing, the self-attention network prompts the robot to focus on information indicative of potential collision hazards, thereby enhancing its obstacle avoidance capabilities.

C. DRL-based Mobile Robot Navigation

In order to verify the effectiveness of the proposed methods on the robot navigation task, a reinforcement learning framework for the navigation of mobile robots was designed, as illustrated in Fig. 2. For the DRL method, the decision was made to employ the SAC, an algorithm that has demonstrated notable efficacy in the domain of robot control tasks. The DRL framework is designed to utilize a feature extraction network to obtain environmental information. The feature extraction results are then input into the actor and critic networks.

The feature extraction network under consideration is a Multi-Layer Perceptron (MLP) network composed of seven fully connected layers, with sizes of $[256, 256, 128, 128, 64, 32, 16]$. It is noteworthy that all networks depicted in the figure utilize the Rectified Linear Unit (RELU) as the activation function. The specific settings of the reinforcement learning model are as follows.

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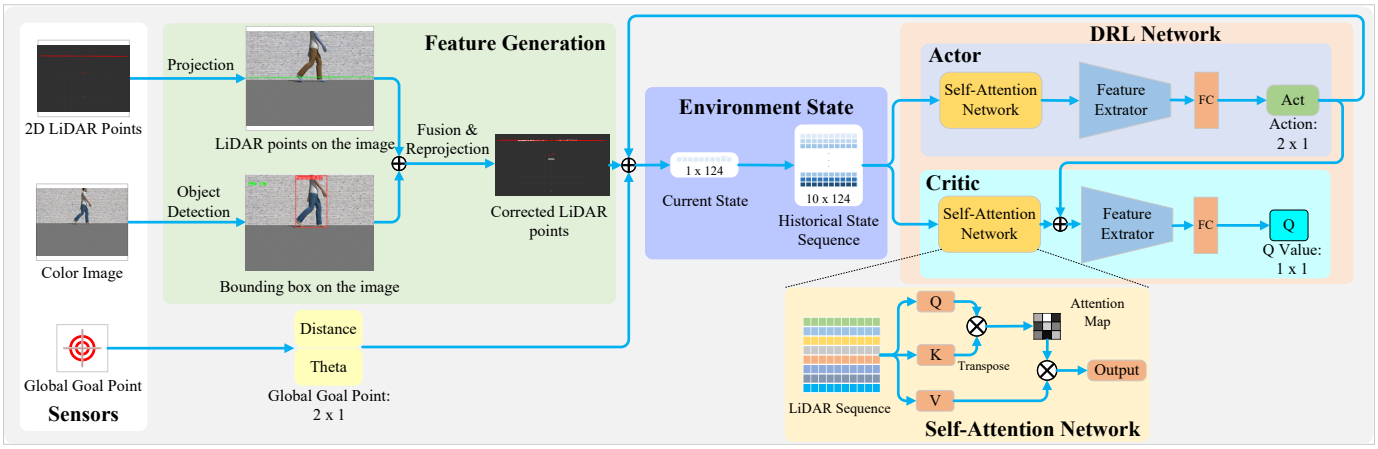

Fig. 2. The DRL framework for mobile robot navigation task.

1) State Space: The state space is comprised of three elements: the corrected LiDAR data, the action output from the previous moment, and the relative position information of the target point. The state of the current moment is denoted by s_t. Thereafter, this current moment state is combined with historical states from the preceding ten moments to become the historical state sequence S_t as

$$\begin{aligned} s_t &= [L_{current}, a_{last}, D_{current}, \theta_{current}], \\ S_t &= \{s_t, s_{t-1}, s_{t-2}, ..., s_{t-10}\} \end{aligned} \tag{9}$$

where, the corrected LiDAR data, $L_{current} \in \mathbb{R}_+$, is a 120-dimensional dataset that captures a 180-degree range in front of the robot. The action output from the previous moment, $a_{last} \in [-1, 1]$, is represented as a 2D vector, including forward and steering outputs. The relative position information of the goal point, $D_{current} \in \mathbb{R}_+$ and $\theta_{current} \in [-\pi, \pi]$, is also a 2D representation in polar coordinates, describing the target's distance and angle relative to the robot. Subsequent combination of these elements results in a 124-dimensional state, which represents the current moment.

2) Action Space: In accordance with the characteristics of the mobile robot, two actions are employed to regulate its forwarding and steering, respectively. The action space encompasses two dimensions, designated as a_0 for the throttle and a_1 for the steering angle. These dimensions are constrained within the interval [-1, 1]. Prior to the implementation of the action within the environment, it is imperative to ascertain that a_0 is always greater than 0 to ensure optimal training conditions. Subsequently, the action should be scaled by the maximum linear velocity v_{max} and the maximum angular velocity ω_{max} as

$$a = [v_{max}(\frac{a_0 + 1}{2}), \omega_{max} a_1] \tag{10}$$

3) Reward Function: The reward function that has been meticulously designed for the robot navigation task in this paper consists of three items. The parameters of each reward have been specifically calibrated to align with the parameters of the experimental scenario. At each time step, a corresponding reward value is assigned as

$$R = \begin{cases} R_{goal} & \text{if goal} \\ R_{collition} & \text{if collition} \\ R_p & \text{else} \end{cases} \tag{11}$$

where, R_{goal} is the reward for reaching the goal point, set to 100; $R_{collition}$ is the reward for a collision, which is set to -100; R_p denotes the reward received during the robot's movement. The reward is composed of five sub-items, and it is calculated based on the current state

$$R_p = r_{straight} + r_{close} + r_{slow} + r_{smooth} + r_{dis} \tag{12}$$

where,

$$\begin{aligned} r_{straight} &= 2a_0 - |a_1| \\ r_{close} &= \begin{cases} -(0.5 - l_{min}) & \text{if } l_{min} < 0.5 \\ 0 & \text{else} \end{cases} \\ r_{slow} &= \begin{cases} -(0.1 - a_0) & \text{if } 0 < a_0 < 0.05 \\ 0 & \text{else} \end{cases} \\ r_{smooth} &= \begin{cases} -1 & \text{if } \left|a_1 - a_1^{old}\right| > 0.3 \\ 0 & \text{else} \end{cases} \\ r_{dis} &= \frac{1}{D_{current}} \end{aligned} \tag{13}$$

In R_p, $r_{straight} \in \mathbb{R}$ denotes the disparity between the throttle and steering angles, prompting the robot to accelerate while maintaining a reduced steering angle. The variable $r_{close} \in \mathbb{R}$ represents a penalty term for close proximity to an obstacle, with a corresponding negative value assigned when the LiDAR reading falls below 0.5 meters. $r_{slow} \in \mathbb{R}$ signifies a penalty term for low speed. When the robot's forward speed is inadequate, a negative value will be assigned. $r_{smooth} \in \mathbb{R}$ is a penalty for sudden changes in the robot's steering angle. It has a fixed negative value and is applied when the difference between the current and previous steering angles exceeds a certain threshold. $r_{dis} \in \mathbb{R}$ is a goal-driven reward. The

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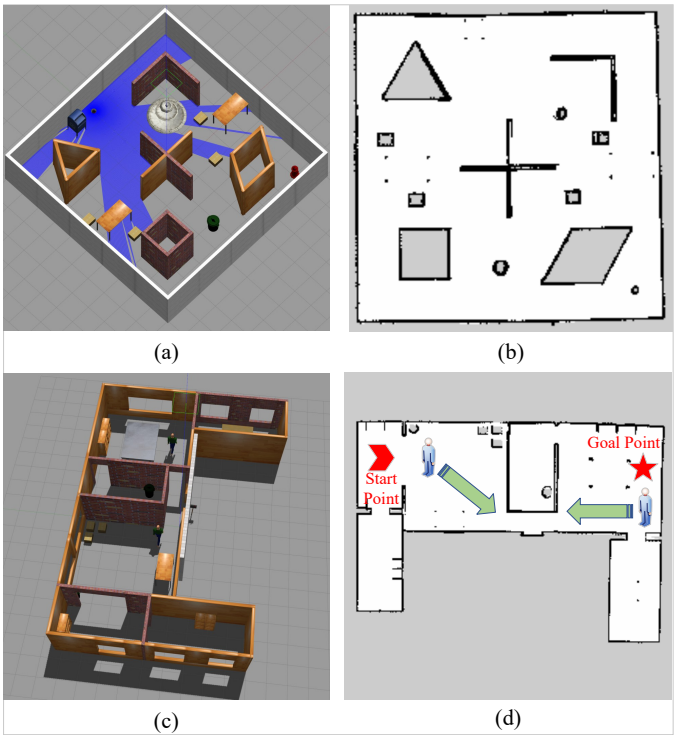

Fig. 3. The simulation scenarios created in Gazebo. (a) The training scenario in Gazebo. (b) Map of the training scenario. (c) The indoor scenario. (d) Map of the indoor scenario.

smaller the value of the current distance from the goal point, the greater the reward.

IV. Experimentation

In order to demonstrate the effectiveness and performance of the proposed algorithm, a series of 3D simulation experiments were conducted using the Gazebo simulator. This section will present an overview of the experimental settings, the training process, and the experiment results.

A. Experimental Setup

Due to the inherent characteristics inherent to RL, robot training is carried out in simulation scenarios. In this paper, the Gazebo software is employed to construct the simulation environment and the robot model. The ROS System is utilized to facilitate the transmission and reception of control instructions and sensor data. The training scenario is illustrated in Fig. 3(a)(b). To conduct simulation tests, TurtleBot3 robots equipped with a webcam and LiDAR sensors were used. The maximum speed of TurtleBot3 was constrained to 0.22 m/s. A webcam with a field of view of $100°$ was employed, while the LiDAR had a measurement range set to $[0.15, 3.5]$ m. This LiDAR scanner covers a complete scanning range of $360°$ with an angular resolution of $0.5°$.

B. Training procedure

During the training process, an episode is considered completed when the robot reaches the endpoint, collides with the wall, or exceeds the maximum time permitted per round. A total of 5000 episodes are trained, and the rewards obtained in each episode are recorded. After every 50 episodes of training, 20 epochs of testing are performed using only the mean of the Gaussian-distributed actions output from the actor network. In the event that these 20 tests successfully reach the goal point, the current neural network weights are saved.

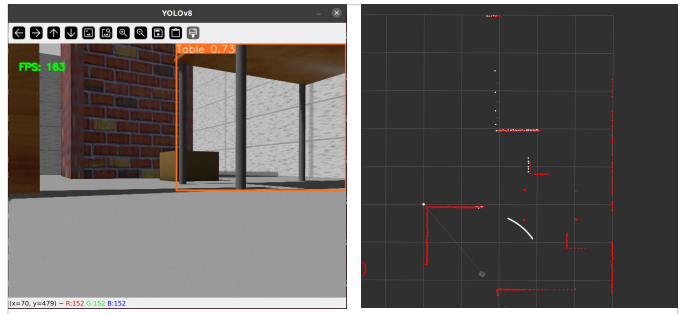

Fig. 4. Demonstration of the fusion effect of LiDAR and camera. (a) The robot utilizes the YOLOv8 model to identify obstacles in the training scenario. (b) LiDAR scanning of the current environment before and after correction.

C. Validation of the sensor information fusion

Prior to conducting navigation experiments, the effectiveness of the proposed sensor fusion method was validated. In the context of the application scenarios of small mobile robots, the selection of obstacles is primarily focused on common indoor obstacles, including pedestrians, tables, chairs, and similar furniture. The corresponding models were imported into the Gazebo simulation environment, and image data was collected using the robot's camera sensor. A training dataset was created for the purpose of training a YOLOv8 model.

Subsequently, the LiDAR data were corrected using the obtained bounding box obtained by the YOLOv8 model, and the results are presented in Fig. 4. The red point cloud represents the original LiDAR data, while the white point cloud illustrates the corrected LiDAR data. The corrected LiDAR data accurately captures the overall dimensions of nearby obstacles, enhancing the robot's ability to recognize obstacles in its environment.

D. Testing in Simulation Scenarios

To assess the efficacy of the integrated sensor fusion approach and the self-attention network, navigation experiments were conducted in two distinct scenarios.

- The straight corridor scenario is a simulation of a simple corridor in which a succession of pedestrians approach from a distance directly in the middle, as shown in Fig. 5(a).
- The indoor scenario is designed to replicate the interior of a standard residential structure, wherein the robot is required to traverse the entirety of the space through narrow corridors. The scenario incorporates two simulated pedestrians, which moves in a reciprocal manner. The intricate design of this scenario is delineated in Fig. 3(c)(d).

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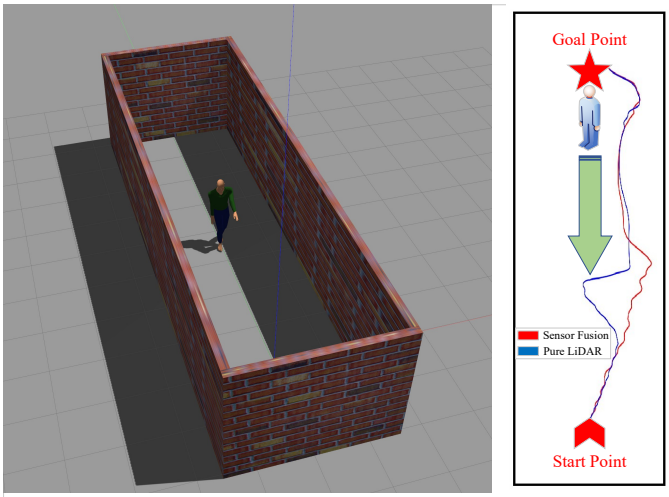

(a) (b)

Fig. 5. The straight corridor scenario and the trajectory obtained therein. (a) The straight corridor scenario in Gazebo. (b) The trajectory obtained by robot navigation using different perceptual inputs.

TABLE I
COMPARISON OF TEST RESULTS FROM VARIOUS METHODS IN THE INDOOR SCENARIO.

Algorithm	Obstacle scenario (30 rounds)			
	Numbers of failure	Time spend (s)	Moving distance (m)	Average speed (m/s)
TD3-ATT [14]	22	109.26	**13.74**	0.13
SAC	9	**74.88**	15.05	**0.20**
SAC-SA(LiDAR)	15	76.74	14.57	0.19
SAC-SA(Fusion)	**4**	76.92	14.55	0.19

In the straight corridor scenario, the navigation effects of using pure LiDAR and sensor fusion methods as sensory inputs are compared, and the typical trajectories obtained are shown in Fig. 5(b). It is evident that robots reliant on LiDAR for perception encounter challenges in accurately detecting dynamic obstacles that undergo shape changes. If only a small area on two legs of pedestrians is scanned, they can easily be ignored, leading to collisions. In contrast, robots that employ sensor fusion to perceive their environment can achieve consistent correction results, provided they are capable of recognizing pedestrians. This capability enables frequent recognition of potential obstacles, thereby facilitating the prevention of collisions.

In the indoor scenario, a series of ablation experiments were conducted and subsequently compared with a recently developed DRL-based navigation method. A total of 100 rounds of tests were conducted, and the number of failures, time spent, and moving distance of each method were recorded. The specific test results are shown in table I, where SAC denotes the version that only uses the sensor fusion method, and SAC-SA denotes the version with the addition of a self-attention network. The version incorporating both the sensor fusion method and the self-attention network exhibits the highest navigation success rate, thereby indicating that the proposed method facilitates robot's discernment and circumvention of obstacles within a reinforcement learning navigation framework.

V. CONCLUSION

This paper proposes a multi-sensor information fusion method that aims to enhance the perception and expression ability of LiDAR by leveraging image information. Subsequently, the perception of key environmental elements is augmented by integrating a self-attention network. To assess the impact of this enhanced perception on robot navigation based on DRL, a simulation environment and a navigation framework are designed. The simulation outcomes, derived from the Gazebo simulator, substantiate the efficacy of the proposed sensing scheme. Subsequent endeavors will entail the evaluation of the proposed method's practical application in real-world environments and its implementation in robot navigation tasks. Furthermore, efforts will be made to enhance the deep reinforcement learning algorithm.

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